
TrustGraspNet: Trust-Aware IMU-Only Grasp Assessment for Soft Spiral Robots under Sparse Sensing

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Abstract

Soft spiral robots can grasp fragile and irregular objects with remarkable adaptability, but reliable grasp assessment remains difficult because grasp state is distributed across compliant body deformation rather than concentrated at a few rigid contacts. We present TrustGraspNet, a compact trust-aware multi-task model that predicts grasp success, grasp quality, intervention action, object family, object size, and per-IMU reliability from sparse onboard proprioception alone. The method uses eight D-optimal IMUs and tendon-displacement commands, combines shared per-sensor encoding with validity-aware fusion and a lightweight Conv1D-GRU temporal backbone, and is trained under explicit sensor corruption with per-sensor trust supervision. On 1,000 MuJoCo grasp trials converted into 12,330 labeled windows, the 8-IMU student model achieves 0.9940 ± 0.0009 trial-level AUROC for grasp success, 0.1496 ± 0.0025 quality RMSE, 0.00354 ± 0.00016 size RMSE, and 0.9481 ± 0.0024 window-level action accuracy on clean evaluation. Under fully corrupted evaluation, success remains at 0.9946 ± 0.0015 AUROC and action accuracy at 0.9432 ± 0.0024 , while trust RMSE rises from 0.0477 ± 0.0027 to 0.2093 ± 0.0003 , correctly exposing degraded sensing reliability. We also clarify the scope of the proprioception-only assumption, show how leakage-free trial-level splitting and corruption-aware training reduce overfitting risk, explain the role of held-out trials, and discuss where the model can be used in practice. The results support a clear claim: sparse IMU-only sensing can be sufficient for decision-level grasp assessment when the dominant environment effects are expressed through the robot’s own deformation dynamics.

1 Introduction

Soft spiral robots are attractive manipulators because their compliant bodies can wrap around fragile, irregular, and uncertain objects without relying on the tightly localized contacts typical of rigid grippers [1–4]. The same compliance that gives soft robots their versatility also makes feedback difficult. In a rigid hand, grasp state can often be summarized by a small set of contact or force variables. In a soft spiral body, grasp state is instead distributed across deformation, delayed load transfer, enclosure quality, and incipient slip. A useful controller therefore needs more than a binary success label: it needs an early estimate of whether the grasp will hold, whether current sensor streams are trustworthy, and whether the robot should continue, tighten, retry, or abort.

Most existing grasp-assessment pipelines obtain this information from external perception or rich contact sensing. Vision, tactile arrays, and multimodal sensing have produced strong results in grasp prediction and manipulation, but they also introduce calibration burden, wiring complexity, packaging overhead, and additional computation [5, 6, 8–10]. These costs are especially problematic in compact soft platforms, where sensing hardware can quickly dominate the design itself. Proprioception is appealing because it is inherently onboard and naturally aligned with closed-loop control. Among practical proprioceptive modalities, IMUs are especially attractive: they are inexpensive, compact, and off-the-shelf. However, prior IMU-based work in soft robotics has focused mainly on shape estimation, state reconstruction, and feedback control rather than decision-level grasp assessment [11–14]. Moreover, IMU degradation such as noise, drift, freezing, delay, dropout, or spoofing is usually treated as a nuisance to be filtered away, not as uncertainty that should itself be estimated.

This project asks a deliberately focused question: *can sparse proprioception alone support reliable decision-level grasp assessment under realistic sensor corruption?* The answer matters scientifically and practically. If the answer is negative, richer sensing is justified. If it is positive, sparse sensing becomes a principled design target rather than a compromise.

Our contributions are fourfold. First, we formulate IMU-only grasp assessment for a soft spiral robot as a trust-aware multi-task sequence-learning problem with outputs for success, quality, action, family, size, and per-sensor trust. Second, we combine D-optimal sensor placement with sparse validity-aware temporal fusion to preserve grasp-relevant information under an 8-IMU hardware budget. Third, we provide five-fold evidence that sparse proprioception is sufficient for accurate grasp assessment and that corruption-aware training is the dominant source of robust task performance. Fourth, we explicitly address questions that are central for a course-project report but often left implicit in papers: when the proprioception-only assumption holds, how overfitting is checked, what held-out trials differ from training, and where the system can realistically be deployed.

2 Related Work and Positioning

Soft robotic grasping and feedback. Soft grippers exploit compliance to handle unknown, delicate, and irregular objects, but this same compliance makes grasp state difficult to observe and regulate [5–7]. Recent systems therefore integrate tactile, force, optical, and multimodal sensing to estimate slip, force, and object properties [9, 10]. These approaches are effective, yet they often rely on specialized fabrication, dense sensor routing, or heavier signal-processing stacks that are difficult to scale to lightweight soft manipulators.

Proprioception in soft robots. A parallel literature studies proprioception through embedded strain sensing, deformable sensors, or inertial sensing distributed along the body [11, 12]. IMUs are particularly attractive because they are compact, inexpensive, and easy to integrate. Existing IMU-based methods have shown promise for soft-body reconstruction and feedback control [13, 14], but they do not directly solve the decision problem considered here: predicting from short windows of onboard dynamics whether a grasp will succeed and what corrective action should be taken.

Learning-based grasp assessment and edge deployment. Vision and tactile learning pipelines have demonstrated strong grasp-outcome prediction, but they depend on richer sensing and heavier compute [15–17]. In parallel, quantization and model compression have enabled compact edge models for robotics and embedded ML [18–20]. These two strands have rarely been combined in soft robotic grasp assessment, where the emphasis is often either rich perception or compliant hardware rather than sparse, deployable, corruption-aware inference.

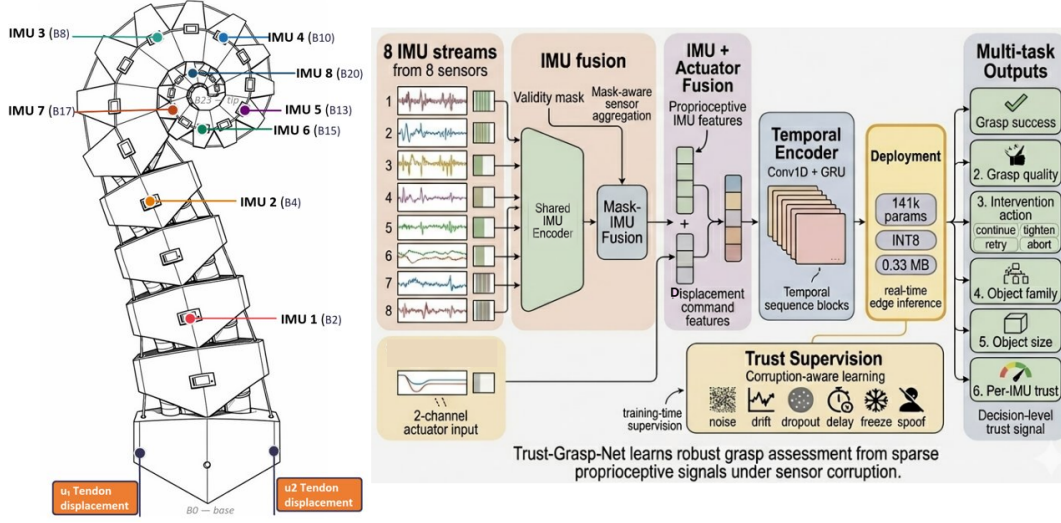


Figure 1: Overview of TrustGraspNet. Eight D-optimal IMUs are mounted along the SpiRob backbone and fused with two tendon-displacement commands. The model uses shared IMU encoding, mask-aware fusion, a lightweight temporal encoder, and explicit trust supervision to predict success, quality, intervention action, object family, object size, and per-IMU trust from sparse proprioception alone.

How our method differs. Relative to prior soft-robot proprioception work, our method targets *decision-level grasp outcomes* rather than shape reconstruction or low-level state estimation. Relative to tactile- and vision-heavy grasping pipelines, it uses *only onboard IMUs and actuator commands*, which is important when sensing, wiring, and compute are all constrained. Relative to generic embedded ML papers, it adds an explicit *per-sensor trust objective under realistic corruption*. This positioning is the core significance of the project: the work is not simply another compact network, but a study of whether sparse body dynamics can replace heavier sensing for robust grasp assessment.

3 Method

3.1 Problem Formulation and Labels

We consider a tendon-actuated soft spiral robot instrumented with M IMUs. At time step t , IMU m provides angular velocity and linear acceleration,

$$\mathbf{x}_{t,m} = [\omega_{t,m}^x, \omega_{t,m}^y, \omega_{t,m}^z, a_{t,m}^x, a_{t,m}^y, a_{t,m}^z] \in \mathbb{R}^6, \quad (1)$$

and the actuator input is the tendon-displacement command

$$\mathbf{u}_t = [u_t^{(1)}, u_t^{(2)}] \in [0, 1]^2. \quad (2)$$

A binary validity mask $\mu_{t,m} \in \{0, 1\}$ indicates whether sensor m is available at time t . The model operates on a temporal window of length $T = 80$ (0.40 s at 200 Hz), yielding

$$\mathbf{X} \in \mathbb{R}^{T \times M \times 6}, \quad \mathbf{U} \in \mathbb{R}^{T \times 2}, \quad \boldsymbol{\mu} \in \{0, 1\}^{T \times M}. \quad (3)$$

In the final sparse-sensing setting, $M = 8$.

Each simulated trial follows the nominal phases *pack*, *reach*, *wrap*, *tighten*, and *pull*. Trial-level supervision is computed from attachment ratio r_{att} , contact ratio r_{con} , lift height Δz , pull-force proxy F_{pull} , and a slip penalty s_{slip} . The continuous grasp-quality target is

$$q = \text{clip}\left(0.45r_{\text{att}} + 0.20r_{\text{con}} + 0.15 \text{clip}(\Delta z/0.05, 0, 1) + 0.10 \tanh(F_{\text{pull}}/4) + 0.10(1 - s_{\text{slip}}), 0, 1\right), \quad (4)$$

and binary success is defined conservatively as

$$y_{\text{succ}} = \mathbb{I}[r_{\text{att}} \geq 0.45 \wedge r_{\text{con}} \geq 0.10 \wedge q \geq 0.45]. \quad (5)$$

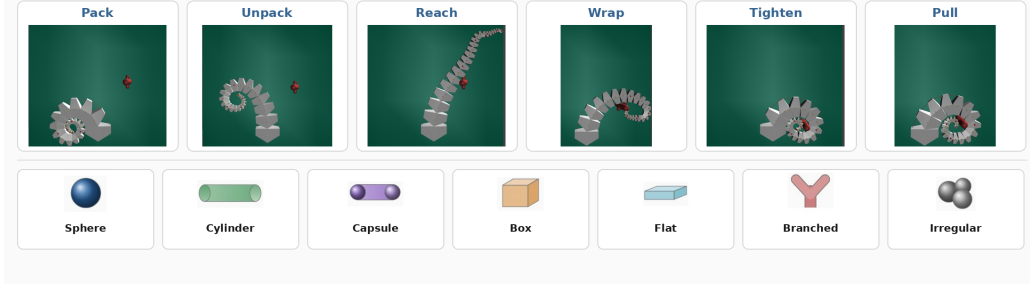


Figure 2: Simulation protocol used for dataset generation. Top: representative phases from pack to pull. Bottom: the seven object families used in simulation. The dataset intentionally spans both easy geometric primitives and harder non-convex morphologies.

Each full trial is segmented into overlapping windows of length 80 and stride 20. Every window inherits trial-level labels for success, quality, object family, and size, and receives an oracle action label from a simple rule-based policy that outputs *continue*, *tighten*, *retry*, or *abort*.

3.2 D-Optimal IMU Placement

Because the physical platform can mount only a small number of IMUs, sensor locations are selected by D-optimal subset design. Let $\Theta \in \mathbb{R}^{N \times P}$ be the matrix of centered inter-segment angle snapshots collected from simulation, where P is the number of candidate sensor locations. Principal component analysis yields the dominant shape basis $\Phi \in \mathbb{R}^{P \times k}$. For a candidate subset S , the placement score is

$$J(S) = \log \det (\Phi_S^\top \Phi_S / \sigma^2), \quad (6)$$

which favors subsets that preserve as much shape information as possible. The selected 8-IMU subset is fixed for all sparse-sensing experiments.

3.3 Trust-Aware Architecture and Corruption Training

TrustGraspNet combines shared per-sensor encoding, validity-aware fusion, a compact temporal backbone, and a separate trust branch. Each IMU sample is encoded by a shared MLP, validity-aware mean pooling aggregates across sensors at each time step, actuator commands are encoded by a small MLP, and the fused temporal sequence is processed by a lightweight Conv1D-GRU backbone. A window-level representation is then fed to task heads for success, quality, action, family, and size. In parallel, a time-aggregated feature for each IMU is mapped to a trust score $\hat{\tau}_m \in [0, 1]$.

To train robustness explicitly, each clean window is corrupted online with probability 0.5. Corruptions emulate common IMU failure modes: additive noise, drift, clipping, delay, freezing, dropout, and spoofing. For each corrupted sensor, a supervisory trust target $\tau_m^* \in [0, 1]$ is assigned according to its expected residual utility. The network is trained with a weighted multi-task objective,

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{succ}} + \lambda_q \mathcal{L}_{\text{qual}} + \lambda_a \mathcal{L}_{\text{act}} + \lambda_f \mathcal{L}_{\text{fam}} \\ & + \lambda_r \mathcal{L}_{\text{size}} + \lambda_\tau \mathcal{L}_{\text{trust}} + \lambda_{\text{KD}} \mathcal{L}_{\text{KD}}, \end{aligned} \quad (7)$$

where success uses binary cross-entropy, action and family use cross-entropy, and quality, size, and trust use mean-squared error. The deployed student model uses hidden widths 24/64/96 and 140,896 parameters, and it is distilled from a wider teacher (48/128/160) to improve compact-model generalization.

4 Experimental Protocol

The final dataset contains 1,000 simulation trials and 12,330 labeled windows after filtering. Objects come from seven families and vary in size, pose, and workspace position. The standard split is leakage-free at the *trial* level, so overlapping windows from the same trial never appear across train, validation, and test sets. This detail is critical because window-level random splitting would create near-duplicate train/test examples and inflate performance.

Table 1: Dataset and evaluation summary.

Quantity	Value
Simulation trials	1,000
Window length / stride	80 / 20 samples
IMU sampling rate	200 Hz
Total labeled windows	12,330
Train / val / test windows	8,834 / 824 / 1,526
Object families	7
Training corruption probability	50% of windows
Final sensor budget	8 IMUs
Random seeds for final reporting	5

Table 2: Main results for the 8-IMU displacement-only student. Success, family, quality, and size are trial-level; action and trust are window-level.

Metric	Clean	Corrupted
Success AUROC	0.9940 ± 0.0009	0.9946 ± 0.0015
Family accuracy	0.498 ± 0.011	0.492 ± 0.018
Quality RMSE ↓	0.1496 ± 0.0025	0.1521 ± 0.0025
Size RMSE ↓	0.00354 ± 0.00016	0.00353 ± 0.00017
Action accuracy	0.9481 ± 0.0024	0.9432 ± 0.0024
Trust RMSE ↓	0.0477 ± 0.0027	0.2093 ± 0.0003
Parameters	140,896	
INT8 size	0.33 MB	

The evaluation is organized around four questions. Can sparse proprioception support accurate decision-level grasp assessment? Does corruption-aware training improve robust performance under sensor degradation? Is explicit trust supervision necessary for meaningful trust estimates? Does the D-optimal 8-IMU configuration preserve the information available in denser sensing? Success, family, quality, and size are evaluated at the *trial level* by aggregating all window predictions from the same trial before metric computation. Intervention action remains a *window-level* metric because it is phase-local. All main results are reported over five random seeds.

To probe generalization more explicitly, earlier versions of the project also defined two stress-test splits: an *unseen-object* split containing held-out branched and irregular trials, and an *unseen-geometry* split containing larger objects with $r \geq 0.020$. These held-out trials still use the same robot, same controller, and same windowing protocol; what changes is object morphology or geometry, not the task definition itself. This distinction matters when interpreting generalization claims: the evaluation tests transfer across more difficult object regimes within the same simulation environment, not transfer to entirely different physics or hardware.

5 Results

Table 2 shows the central result: the sparse 8-IMU model achieves near-perfect trial-level separation of successful and failed grasps after aggregating overlapping windows within each trial, while also maintaining high action accuracy at the window level. The most important observation is not success AUROC alone, but the stability of the whole task profile under corruption. Success, quality, size, and action shift only modestly between clean and corrupted evaluation, whereas trust RMSE rises sharply. This is exactly the intended behavior: the model remains task-relevant while explicitly signaling that sensing reliability has degraded.

The ablation study sharpens the main claim. Corruption-aware training is the dominant source of task robustness: removing it drops robust action accuracy from 0.9432 to 0.7533 and worsens quality RMSE from 0.1521 to 0.2009. Trust supervision plays a different role. When the trust loss is removed, the task heads remain strong, but trust RMSE becomes poor under both clean and corrupted evaluation. The contribution of trust supervision is therefore not generic performance improvement; it is making the trust signal semantically meaningful.

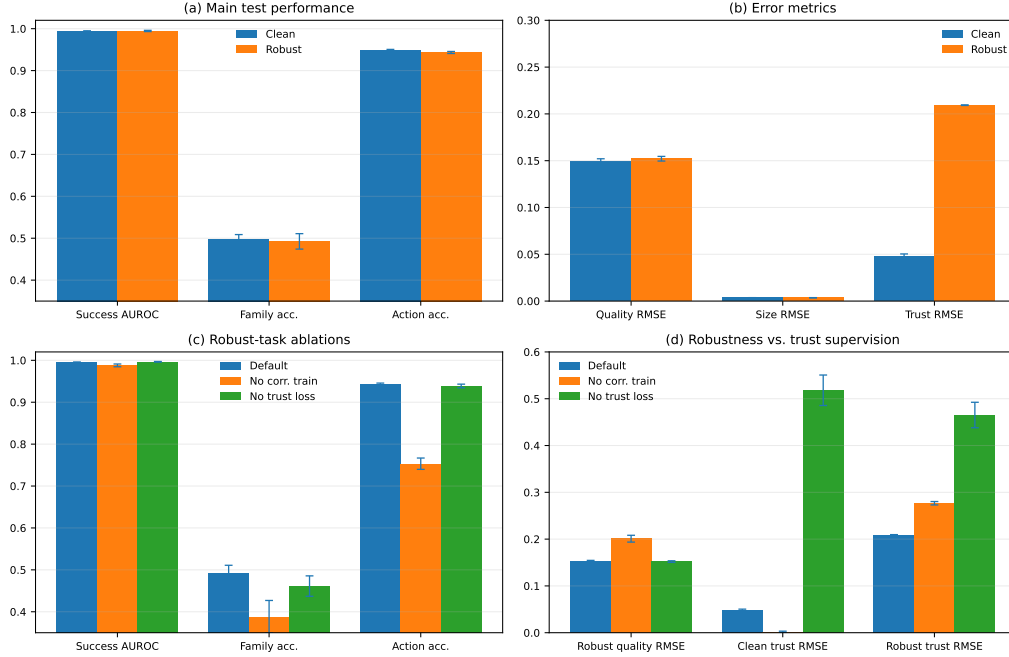


Figure 3: Five-seed robustness summary. Removing corruption-aware training substantially damages robust family accuracy, action accuracy, and quality estimation, whereas removing trust supervision mainly destroys the interpretability of trust estimates.

Sparse design choices are also empirically justified. The D-optimal 8-IMU subset matches a 24-IMU layout within confidence intervals on success, family accuracy, action accuracy, and quality RMSE in the design study, while improving reconstruction-space metrics relative to uniform and hand-picked alternatives. Similarly, replacing the full simulated actuator state with displacement only leaves success and action essentially unchanged, which is important because displacement is the actuator signal that physical hardware can measure most reliably.

6 Discussion: Assumptions, Generalization, and Practical Use

6.1 Does the proprioception-only assumption always hold?

No. The key assumption is not that proprioception is literally independent of the environment in all situations; that would be too strong. The more precise assumption is that the environment influences grasp outcome *mainly through changes in the robot's own deformation and actuation response*, and that these changes are sufficiently consistent to be learned from onboard signals. Under that assumption, IMU streams become an informative proxy for contact state, enclosure quality, and slip tendency.

This assumption holds best when the same robot interacts with objects in a similar operating regime: comparable contact materials, moderate changes in size and pose, quasi-static or mildly dynamic grasping, and disturbances that manifest through the robot body itself. It becomes weaker when important environment factors are poorly reflected in body dynamics, such as strong unmodeled friction changes, impacts from external structures, fluid disturbances, hardware artifacts absent from simulation, or new contact regimes that produce ambiguous deformation signatures. For this reason, the present results should be read as strong evidence within a simulation-first regime, not as proof that IMU-only sensing is universally sufficient.

6.2 How do we check and reduce overfitting?

Several design decisions reduce overfitting risk. First, splitting is done at the trial level rather than the window level, preventing leakage from overlapping windows. Second, online corruption augmentation exposes the model to diverse degraded inputs instead of allowing it to memorize narrow clean-signal patterns. Third, the student network is deliberately compact and uses dropout, AdamW weight decay, and validation-based model selection. Fourth, all main results are averaged across five random seeds rather than reported from a single favorable run. Finally, knowledge distillation from a wider teacher to a smaller student helps retain useful structure without deploying an oversized model.

We also check for overfitting empirically. In earlier training curves, optimization is stable and validation metrics improve and then plateau rather than collapse; the compact model therefore appears to generalize better than a purely larger-capacity alternative. Still, the most convincing future overfitting test is stronger out-of-distribution evaluation on held-out object families, held-out geometries, and real hardware data.

6.3 What differs in held-out trials?

Held-out trials are not random extra windows from the same trajectory. They differ at the *trial-generation level*. In the unseen-object split, branched and irregular objects are held out; in the unseen-geometry split, larger objects are held out. However, they still share core properties with training data: they use the same nominal task, same robot, same controller phases, same sensing modality, and the same window construction. This means the evaluation measures generalization across object morphology and geometry within the same task family, rather than transfer to a wholly different manipulation problem.

6.4 Where can the model be applied, and what gains does it bring?

The most plausible deployment settings are soft grippers operating under sensing, wiring, and latency constraints: lightweight pick-and-place, delicate handling of irregular household items, agricultural or food handling where compliance is helpful, and embedded soft-manipulation systems that cannot rely on external cameras. In these settings, the main gains are practical rather than purely numerical: reduced sensing complexity, fewer cables and custom components, lower onboard compute requirements, and real-time local inference. The current student model fits into roughly 0.33 MB after INT8 quantization and was designed around a 50 ms decision stride, which makes onboard deployment realistic.

The model also brings a more subtle gain: it separates *task prediction* from *sensor reliability estimation*. Even when grasp success remains predictable, a rising trust error or low trust score can tell the downstream controller that the sensing substrate itself has become harder to certify. That is valuable for real robots, where the right action is sometimes not “predict harder” but “act more cautiously.”

7 Conclusion

We presented TrustGraspNet, a compact trust-aware framework for IMU-only grasp assessment in soft spiral robots under sparse sensing. The method combines D-optimal sensor placement, validity-aware temporal fusion, corruption-aware learning, and explicit per-IMU trust estimation in a deployment-oriented student model. The five-seed results show that eight carefully placed IMUs and displacement-only actuation are sufficient for strong trial-level grasp assessment and window-level intervention prediction, that corruption-aware training is the primary driver of robustness, and that explicit trust supervision is necessary for meaningful reliability estimates. At the same time, the paper makes its assumptions explicit: proprioception-only inference works when environment effects are expressed through body dynamics, and it should be stress-tested further under held-out physics and real hardware.

Reproducibility and repository. repository URL. Insert your final private repository link here before submission: PRIVATE_GITHUB_REPO_LINK.

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